

MetaGraphSci – Milestone 2

Graph Transformer Self-Supervised Contrastive Neighbor Adaptation
for Scientific Document Classification

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Problem Identification and Motivation

Scientific document classification is the task of assigning the subject category of a research document based on its content as well as its context within the larger scholarly literature. It is a significant task with applications in digital libraries, semantic search, massive document indexing, and recommendation systems. It is also a challenging task due to the lack of sufficient supervision data, as the classification of a document with respect to a fine-grained classification system requires domain knowledge, and only a small fraction of the documents in the real world are annotated.

A scientific document is a rich medium that offers three sources of information: the *textual signal*, which is the semantic content of the document; the *bibliographic signal*, which includes the publication venue, publisher, authorship, and publication year; and the *relational signal*, which includes the network of citations between documents. A good classifier should not consider the document as just a sequence of words but as a rich object with its context within the larger scholarly literature.

Existing methods only partially exploit this information. Text encoders such as SciBERT (Beltagy et al., 2019) and citation-informed encoders such as SPECTER (Cohan et al., 2020) provide strong semantic representations, but downstream classification is still often driven primarily by text. Graph-based models capture citation dependencies, yet their performance may deteriorate in low-label settings because structural aggregation can blur discriminative semantics, especially when the graph is noisy or weakly homophilous (Rampasek et al., 2022; Wu et al., 2022). Similarly, citation-aware contrastive learning methods improve representation quality through structural supervision, but they usually define similarity only through graph proximity and do not fully exploit metadata as an informative source of relational evidence (Ostendorff et al., 2022; Taechoyotin and Acuna, 2024).

These constraints give rise to the need to develop a semi-supervised pipeline called **MetaGraphSci** that combines text, metadata, and citation context. The main hypothesis is that scientific document classification can be improved by combining semantic encoders with metadata interactions and graph contextualization. The proposed pipeline has three principal components: a scientific text encoder that is pretrained, a metadata interaction encoder, and a citation context transformer. The training is working in two-stage: a metadata-informed contrastive pretraining with neighborhood awareness, followed by adaptive pseudo-label fine-tuning in scenarios with limited labeled data.

Scientific Document Representation from Text

One of the primary methodologies that this field focuses on is the use of pre-trained models that are trained on scientific text data. An example of this would be the use of the SciBERT model (Beltagy et al., 2019), which makes use of the vocabulary and corpus that are specific to the domain of scientific discourse. However, the SciBERT model remains at the top of the list for general document representations in the scientific domain. Furthermore, the SPECTER model (Cohan et al., 2020) builds on this approach by taking into consideration the citation information during the pre-training phase, where it learns to put papers with similar citation patterns close together in the embedding space, which remains a major improvement over the existing state of the art in citation-aware document representations.

SciNCL (Ostendorff et al., 2022) continued to improve upon the effectiveness of the contrastive learning approach by sampling from the K-hop citation neighborhood of each paper instead of relying on co-citation relationships. This work is of particular relevance to the proposed model as it highlights the importance of neighborhood-aware contrastive learning as opposed to batch-aware contrastive learning. Also recent works propose that scientific documents should not simply be treated as text sequences. CoSAEmb (Singh and Singh, 2024) proposed section-aware contrastive representations of scientific documents, while MISTI (Taechoyotin and Acuna, 2024) showed the effectiveness of incorporating metadata into the contrastive learning approach. This indicates that more effective representations of scientific documents can be achieved by making use of structure beyond the title and abstract of a paper.

Graph Neural Networks and Graph Transformers

Citation graphs provide a natural foundation for modeling relationships between scholarly documents, and graph neural networks have therefore become a standard approach for node-level tasks in this domain. But, traditional message-passing architectures exhibit notable limitations. In particular, repeated local aggregation tends to homogenize node representations, leading to the well-known problem of over-smoothing as network depth increases. This effect is especially detrimental in low-label settings, where preserving semantic discriminability is critical.

GraphGPS (Rampasek et al., 2022) addresses this issue by combining local message passing with global attention, resulting in a hybrid architecture that balances expressiveness with scalability. Similarly, NodeFormer (Wu et al., 2022) demonstrates that transformer-based reasoning on graphs can be made efficient through learned structure induction and attention approximations. More broadly, the development of graph transformers with explicit structural priors has been explored in Yun et al. (2019) and further generalized by Dwivedi and Bresson (2022). These works highlight the importance of incorporating structural encodings, such as topological and spectral features, to enhance transformer performance on graph-structured data. This perspective directly informs the design of the citation-context encoder in MetaGraphSci.

Semi-Supervised Learning and Pseudo-Labeling

Because offered labels are scarce in scientific corpora, semi-supervised learning is central to the problem. Weakly supervised methods such as that of Meng et al. (2020) showed that useful classifiers can be bootstrapped even when explicit annotations are limited, highlighting the value of unlabeled data when handled carefully.

More recent work has focused on pseudo-label selection. [Yang et al. \(2025\)](#) proposed class-adaptive pseudo-label thresholds, showing that fixed global thresholds bias selection toward dominant classes and exacerbate class imbalance. Similarly, [Chuang et al. \(2024\)](#) showed that monitoring pseudo-labeling status across epochs can reduce confirmation bias and improve sample selection. These works are directly relevant to the second training stage of MetaGraphSci, where reliable pseudo-label generation is required under imbalance and uncertainty.

Fine-Grained Scholarly Classification Benchmarks

For fine-grained scientific classification, it is necessary that the data reflects the semantic diversity of the information contained in the data, as well as the inherent class imbalance that is present in the data, which reflects the real world. FoRC4CL ([Ahmad et al., 2024a](#)) is particularly relevant in this regard, as it provides a manually curated data set of NLP papers, classified according to a fine-grained field of research taxonomy. The analyses that follow ([Ahmad et al., 2024b](#); [Dampa, 2025](#); [Francis et al., 2025](#)) reveal that hierarchical interdependencies between classes, as well as class imbalance, are key concerns that persist with regard to the actual classification task, making the data set useful not only for evaluating the accuracy of the model, but also its ability to retain discriminative ability under real-world, intricately structured conditions.

Best Existing Datasets and Solutions

Datasets

In this study, FoRC4CL ([Ahmad et al., 2024a](#)) is seen as a prominent evaluation resource for this specific problem, considering that it is designed to classify NLP articles at a detailed level. The hierarchical structure and class imbalance properties of this specific benchmark make it particularly useful as a resource to evaluate the robustness of a system under practical circumstances. Finally, with regards to citation network benchmarks such as **Cora**, **PubMed**, and **ogbn-arxiv** it is noted that they remain applicable within the context of graph node classification problems. Even though they are not as specific as FoRC4CL, they remain appropriate controlled environments to compare various structural modeling techniques with each other, as well as to test their scalability.

Best Existing Solutions

Among text-and citation-aware encoders, SciNCL ([Ostendorff et al., 2022](#)), as one of the state-of-the-art models in this space, is particularly prominent in this setting due to the combination of scientific language modeling with neighborhood-aware contrastive supervision. The development context of SciNCL is particularly relevant given the more nuanced consideration of meaningful positive context as opposed to citation-aware context.

Among graph-based models, the balance of local relational information and global transformer attention in GraphGPS ([Rampasek et al., 2022](#)) is particularly robust, as is the scalable transformer variant of NodeFormer ([Wu et al., 2022](#)). These models are particularly relevant as structural baselines to MetaGraphSci. In the space of semi-supervised optimization, the pseudo-labeling strategy based on class adaptivity proposed by [Yang et al. \(2025\)](#) is particularly pertinent to the second training stage of this proposed idea. While not originally motivated by citation graph document classification, the underlying principle is particularly relevant to this low-label scholarly context.

Comparative Analysis and Proposed Improvements

Although the above methods have substantially advanced scientific document representation learning, several limitations remain open.

First, **metadata is still underutilized**. Methods such as SPECTER and SciNCL define positive relations primarily through citation structure. However, in real scholarly corpora two documents may be strongly related without citing one another directly, for example because they share venue, publication period, or authorship patterns. This motivates the introduction of an explicit metadata affinity function $\mu(i, j)$, which refines the notion of document similarity beyond the citation graph alone.

Second, **graph transformers rarely model bibliographic compatibility explicitly**. Existing graph-transformer architectures generally rely on node features and structural encodings, but do not directly incorporate metadata as a bias in attention. In scientific corpora, however, bibliographic compatibility often acts as an informative relational prior. Injecting such compatibility into attention is therefore a natural extension for scholarly document modeling.

Third, **pseudo-labeling is usually treated as an instance-wise decision problem**. Standard pseudo-label mechanisms rely only on confidence estimates and do not explicitly account for the fact that graph neighbors may occupy semantically related regions of the label space. This suggests that graph-aware calibration and selection are especially important in citation-based corpora.

Limitations of Existing Methods	Improvements in MetaGraphSci
Text-only or weakly graph-aware models do not jointly exploit semantic, structural, and bibliographic evidence.	Unified multimodal modeling over text, citation context, and metadata.
Citation-aware contrastive learning usually defines similarity only from graph proximity.	Metadata-aware similarity through an explicit affinity function $\mu(i, j)$.
Graph-transformer attention is typically independent of bibliographic compatibility.	Metadata-conditioned attention bias integrated into the citation-context encoder.
Contrastive learning may treat graph-neighbor documents as negatives, creating false-negative pressure.	Neighborhood-aware masking of the negative pool during contrastive pretraining.
Pseudo-label selection often relies on fixed or weakly adaptive thresholds.	Class-adaptive pseudo-labeling with confidence calibration and dynamic thresholds.

Table 1: Comparison between existing limitations and the design goals of MetaGraphSci.

Identified Alternatives

Some of these alternatives were considered during the design of MetaGraphSci.

For the text encoder, a citation-aware model such as SPECTER (Cohan et al., 2020) could be used as the primary backbone. However, SciBERT remains a strong and flexible choice, particularly when citation structure is modeled explicitly during contextualization rather than fully embedded in pretraining.

For the graph encoder, GraphGPS (Rampasek et al., 2022) and NodeFormer (Wu et al., 2022) provide competitive solutions with different trade-offs in scalability and expressiveness.

In contrast, our approach focuses on directly integrating bibliographic compatibility into graph reasoning, instead of treating structural and metadata signals separately.

Regarding the training objective, purely citation-based contrastive learning, as in SciNCL (Ostendorff et al., 2022), represents a strong baseline. We extend this idea by incorporating metadata-based compatibility, enabling a more comprehensive notion of similarity aligned with the multimodal nature of scientific documents.

Finally, while standard pseudo-labeling methods (Meng et al., 2020; Chuang et al., 2024; Yang et al., 2025) can be applied independently, we adopt a different perspective: pseudo-labeling should be tightly coupled with the learned multimodal representation, leveraging text, graph structure, and metadata jointly. Overall, MetaGraphSci is based on the premise that scientific documents are not isolated text units, but structured entities embedded in citation networks and enriched by bibliographic information. Effective classification therefore requires integrating these signals in a unified manner. The code for MetaGraphSci is publicly available at <https://github.com/justin-marian/metagraphsci/>.

Proposed Solution

MetaGraphSci encodes each document through three complementary branches: a pretrained scientific language model for textual semantics, a Deep & Cross metadata encoder for bibliographic attributes, and a Graph Transformer topology encoder for structural and relational context. The three representations are combined through a gated multimodal fusion mechanism:

$$\begin{aligned} u_i &= \text{MLP}_{\text{fuse}}([h_i^{(t)} \oplus h_i^{(m)} \oplus h_i^{(c)}]) \\ g_i &= \sigma(\mathbf{W}_g[h_i^{(t)} \oplus h_i^{(m)} \oplus h_i^{(c)}] + b_g) \\ z_i &= g_i \odot u_i \end{aligned} \quad (1)$$

Here u_i is the shared multimodal projection, g_i is an input-dependent gating coefficient, and z_i is the final latent representation. The gated fusion allows the model to learn, per document, how much to rely on each modality—text, metadata, or citation structure—depending on which signals are most informative for that paper.

Text Encoder

The text encoder maps the title and abstract of document i to a semantic representation using a pretrained scientific language model. The input sequence is defined as:

$$x_i = [\text{CLS}] \oplus \text{title}_i \oplus [\text{SEP}] \oplus \text{abstract}_i \oplus [\text{SEP}] \quad (2)$$

A pretrained SciBERT encoder is applied to this sequence. The document embedding is taken from the final CLS hidden state:

$$\mathbf{H}_i^{(0)} = \mathbf{E}(x_i) + \mathbf{P}, \quad \mathbf{H}_i^{(\ell)} = \text{SciBERT}^{(\ell)}(\mathbf{H}_i^{(\ell-1)}), \quad \ell = 1, \dots, L, \quad h_i^{(t)} = \mathbf{H}_i^{(L)}[0] \in \mathbb{R}^{d_t} \quad (3)$$

Metadata Encoder

The metadata encoder processes bibliographic attributes through a Deep & Cross Network. Categorical fields (venue, publisher) are embedded into trainable vectors; numerical fields (year) are

normalized and projected; author embeddings are mean-pooled. The input metadata descriptor:

$$u_i^{(m)} = \left[\bigoplus_{a \in \mathcal{C}} e_i^{(a)} \oplus \bigoplus_{b \in \mathcal{R}} r_i^{(b)} \right] \quad (4)$$

Cross network iteratively introduces higher-order feature interactions. For $\ell = 0, \dots, L_m - 1$, the final metadata embedding is:

$$x_i^{(\ell+1)} = x_i^{(0)} \left(w_\ell^\top x_i^{(\ell)} \right) + b_\ell + x_i^{(\ell)} \quad h_i^{(m)} = \text{MLP}_m \left(x_i^{(L_m)} \right) \in \mathbb{R}^{d_m} \quad (5)$$

Beyond serving as a modality branch, the metadata embedding plays a second role in the MCNA pretraining objective: it defines a metadata affinity function used to refine the selection of positive pairs (see Section 6.4).

Graph Transformer Topology Encoder

Unlike a standard graph attention network, which computes attention coefficients purely from node feature similarity, the Graph Transformer topology encoder augments the attention mechanism with explicit *structural position encodings* (SPEs). This allows attention to reflect both semantic similarity and graph-topological distance, which is critical for distinguishing topically related but structurally distant papers from genuine citation neighbors.

Structural position encoding. For each document i , a structural position encoding $p_i^{(s)} \in \mathbb{R}^{d_p}$ is computed from the citation graph. Two complementary signals are combined: (i) the normalized Laplacian eigenvectors $\phi_i \in \mathbb{R}^k$, which encode global spectral position, and (ii) the K -hop distance profile $d_i \in \mathbb{R}^K$, whose k -th entry counts the number of documents at hop distance k . These are projected and summed:

$$p_i^{(s)} = \mathbf{W}_\phi \phi_i + \mathbf{W}_d d_i \in \mathbb{R}^{d_p} \quad (6)$$

The text embedding and structural position encoding are combined to form the token that serves as the query and key in the graph attention:

$$\tilde{h}_i = \mathbf{W}_{\text{in}} h_i^{(t)} + p_i^{(s)} \in \mathbb{R}^{d_c} \quad (7)$$

Multi-head graph attention with metadata conditioning. Let $\mathcal{N}(i)$ denote the direct citation neighborhood of document i . For attention head r , the query, key, and value projections of document i and neighbor j are:

$$q_i^{(r)} = \mathbf{W}_Q^{(r)} \tilde{h}_i, \quad k_j^{(r)} = \mathbf{W}_K^{(r)} \tilde{h}_j, \quad v_j^{(r)} = \mathbf{W}_V^{(r)} \tilde{h}_j \quad (8)$$

To incorporate metadata affinity between the query document and its neighbors, we introduce a scalar metadata compatibility bias $b_{ij}^{(m)}$ computed from the dot product of their metadata embeddings:

$$b_{ij}^{(m)} = h_i^{(m)\top} \mathbf{W}_b h_j^{(m)} \quad (9)$$

The attention coefficient for head r is:

$$\alpha_{ij}^{(r)} = \frac{\exp\left(\frac{q_i^{(r)\top} k_j^{(r)}}{\sqrt{d_c/R}} + b_{ij}^{(m)}\right)}{\sum_{k \in \mathcal{N}(i)} \exp\left(\frac{q_i^{(r)\top} k_k^{(r)}}{\sqrt{d_c/R}} + b_{ik}^{(m)}\right)} \quad (10)$$

where R denotes the number of attention heads. The multi-head topology embedding is then:

$$h_i^{(c)} = \sigma\left(\mathbf{W}_O \left[\bigoplus_{r=1}^R \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(r)} v_j^{(r)} \right]\right) \in \mathbb{R}^{d_c} \quad (11)$$

Stacking L_T such layers with residual connections and layer normalization yields the final topology representation. The metadata compatibility bias $b_{ij}^{(m)}$ is the key innovation that distinguishes this encoder from prior Graph Transformers: it ensures that documents sharing venue, publication period, or authorship attract higher attention weight even when their text embeddings are less similar, directly exploiting the structured metadata signal during topology encoding.

Meta-Informed Contrastive Neighbor Adaptation

The pretraining objective drives the encoder toward representations where structurally and bibliographically related documents are close in latent space. We define positive pairs through a joint criterion that requires both citation proximity and metadata affinity.

Metadata affinity. Two documents i and j are metadata-compatible, written $\mu(i, j) = 1$, if they share at least one categorical metadata attribute (e.g., the same venue, overlapping authors, or publication within a two-year window). Formally:

$$\mu(i, j) = \mathbf{1}[\text{venue}(i) = \text{venue}(j) \vee \mathcal{A}(i) \cap \mathcal{A}(j) \neq \emptyset \vee |y_i - y_j| \leq 2] \quad (12)$$

where $\mathcal{A}(i)$ denotes the author set of document i and y_i denotes its publication year.

Positive pair construction. For an anchor document i , a positive view i^+ is sampled from documents in $\tilde{\mathcal{N}}^K(i)$ that also satisfy $\mu(i, i^+) = 1$. If no metadata-compatible neighbor exists, a standard data-augmented view (token dropout, synonym replacement) is used as a fallback. This policy ensures that the contrastive signal reinforces both topological and bibliographic coherence.

Debiased negative sampling. Documents in $\tilde{\mathcal{N}}^K(i)$ are excluded from the negative set of anchor i . Additionally, documents satisfying $\mu(i, j) = 1$ are soft-downweighted in the negative pool via a penalty weight $\rho \in [0, 1)$. The meta-informed contrastive neighbor adaptation (MCNA) loss is:

$$\mathcal{L}_{\text{MCNA}} = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \log \frac{\exp(\text{sim}(z_i, z_{i^+})/\tau)}{\exp(\text{sim}(z_i, z_{i^+})/\tau) + \sum_{j \in \mathcal{N}^-(i)} w_{ij} \exp(\text{sim}(z_i, z_j)/\tau)} \quad (13)$$

where $\mathcal{N}^-(i) = \mathcal{B} \setminus \{i\} \setminus \tilde{\mathcal{N}}^K(i)$ is the debiased negative pool and the per-pair weight is:

$$w_{ij} = \begin{cases} \rho & \text{if } \mu(i, j) = 1 \\ 1 & \text{otherwise} \end{cases} \quad (14)$$

Compared to standard InfoNCE and to prior neighborhood-aware debiasing approaches, MCNA further removes bibliographic false negatives. This is particularly important in scientific corpora where thematically related papers from the same venue are frequent in any mini-batch but may lack direct citation links.

Class-Adaptive Pseudo-Labeling

During semi-supervised fine-tuning, unlabeled documents whose predictions exceed a per-class confidence threshold are incorporated as pseudo-labeled training examples. Following Yang et al. (2025), the threshold for class c at fine-tuning epoch t is set proportional to the maximum predicted probability for that class across all unlabeled documents:

$$\gamma_c^{(t)} = \beta \max_{j \in \mathcal{D}_U} p_{j,c}^{(t)}, \quad \beta \in (0, 1] \quad (15)$$

The selected pseudo-labeled set is:

$$\mathcal{U}^{(t)} = \left\{ j \in \mathcal{D}_U \mid \max_c p_{j,c} \geq \gamma_{\hat{y}_j}^{(t)}, \hat{y}_j = \arg \max_c p_{j,c} \right\} \quad (16)$$

The semi-supervised objective is the cross-entropy loss over the selected set:

$$\mathcal{L}_{\text{ssl}}(\theta; \mathcal{D}_U, \mathcal{G}) = -\frac{1}{|\mathcal{U}^{(t)}|} \sum_{j \in \mathcal{U}^{(t)}} \log p_{j, \hat{y}_j} \quad (17)$$

By scaling the threshold per class, this formulation avoids the failure mode of a global threshold, which systematically favors high-entropy minority classes or high-confidence majority classes depending on the degree of imbalance. This is a key concern in fine-grained scientific taxonomies where class distributions are highly skewed.

Classification Head

Classification is performed through a set of learnable class vectors $\{w_c\}_{c=1}^C$. Both document representations and class vectors are ℓ_2 -normalized to keep the decision rule consistent with the cosine similarity structure of the contrastive pretraining objective:

$$\bar{z}_i = \frac{z_i}{\|z_i\|_2}, \quad \bar{w}_c = \frac{w_c}{\|w_c\|_2}, \quad o_{i,c} = s \bar{z}_i^\top \bar{w}_c, \quad c = 1, \dots, C \quad (18)$$

The class posterior and final prediction are:

$$p_i = \text{softmax}(o_i), \quad \hat{y}_i = \arg \max_{c \in \mathcal{Y}} p_{i,c} \quad (19)$$

The temperature scale $s > 0$ controls logit sharpness and is learned during fine-tuning.

[t] MetaGraphSci Training Procedure *Two-stage training: meta-informed contrastive neighbor adaptation pretraining followed by class-adaptive pseudo-label semi-supervised fine-tuning.*
 [1] $\mathcal{G}$, $\mathcal{D}_L$, $\mathcal{D}_U$, hop radius K , temperature τ , metadata downweight ρ , confidence scaling β , loss weight λ Trained encoder Φ_θ and class vectors $\{w_c\}_{c=1}^C$ Initialize θ (SciBERT, metadata

encoder, Graph Transformer) and $\{w_c\}$ **Stage 1: MCNA Pretraining** each pretraining epoch Sample mini-batch $\mathcal{B} \subset \mathcal{V}$ Compute SPEs $p_i^{(s)}$ from Laplacian eigenvectors and hop-distance profiles Compute representations $z_i = \Phi_\theta(x_i, m_i, \mathcal{N}(i))$ for $i \in \mathcal{B}$ each anchor $i \in \mathcal{B}$ Sample positive i^+ from $\tilde{\mathcal{N}}^K(i)$ with $\mu(i, i^+) = 1$ (or augmented fallback) Exclude $\tilde{\mathcal{N}}^K(i)$ from negative pool; downweight metadata-compatible negatives by ρ Evaluate $\mathcal{L}_{\text{MCNA}}$ Eq. 13 Update θ by minimizing $\mathcal{L}_{\text{MCNA}}$ **Stage 2: Semi-Supervised Fine-Tuning** each fine-tuning epoch t Sample $\mathcal{B}_L \subset \mathcal{D}_L$ and $\mathcal{B}_U \subset \mathcal{D}_U$ Compute representations and class posteriors for $\mathcal{B}_L \cup \mathcal{B}_U$ Evaluate supervised loss $\mathcal{L}_{\text{sup}}$ on $\mathcal{B}_L$ Compute per-class thresholds $\gamma_c^{(t)}$ and select $\mathcal{U}^{(t)}$ from $\mathcal{B}_U$ Evaluate pseudo-label loss $\mathcal{L}_{\text{ssl}}$ Eq. 17 Update θ and $\{w_c\}$ minimizing $\mathcal{L}_{\text{sup}} + \lambda \mathcal{L}_{\text{ssl}}$ ($\Phi_\theta, \{w_c\}_{c=1}^C$)

Preliminary Results

OpenAlex-AI Dataset Statistics

We used [OpenAlex](#) ([OpenAlex, 2026](#)), an open catalog of scholarly works with extended metadata for scientific papers. For this study, we constructed a dataset focused on the artificial-intelligence subfield within computer science. The collected metadata include document titles and abstracts, authorship information, publication venues and publishers, publication years, citation links, and field or topic assignments, together with related bibliographic attributes such as affiliations and source identifiers. Additional corpus, metadata, temporal, and graph statistics are reported in Table 2.

Characteristic	Value	Characteristic	Value
<i>Volume</i>		<i>Text</i>	
Number of documents	30 000	Avg. title length (words)	8.2
Number of citation edges	244 605	Avg. abstract length (words)	146.5
Nodes in citation graph	27 717	Median abstract length (words)	141
Isolated documents	2 283	<i>Temporal</i>	
<i>Labels</i>		Year range	1858 – 2025
Number of classes	77	Median / IQR year	2010 / [2003, 2018]
Labeling strategy	label_ratio = 0.10	<i>Citation graph</i>	
Distribution	long-tail (min 1, max 2 521)	Average out-degree	8.15
<i>Structured metadata</i>		Evaluation type	transductive, temporal split
Unique venues	3 022		
Unique publishers	606		
Unique authors	60 417		
Avg. authors per paper	3.65		

Table 2: OpenAlex-AI dataset statistics summarizing corpus scale, labels, metadata, text, time span, and citation-graph properties.

OpenAlex-AI Results

The OpenAlex-AI experiments were run on an NVIDIA RTX 5090 instance provisioned through Runpod ([Runpod, 2026](#)), using a Blackwell-class GPU with 32 GB of GDDR7 memory, 21,760 CUDA cores, 1.79 TB/s memory bandwidth, advertised FP16 tensor performance of 0.42 PFLOPS, and PCIe Gen5 $\times 16$ host bandwidth of 63 GB/s. The benchmark used the `openalex_ai` configuration with a temporal transductive split, `label_ratio=0.10`, `test_size=0.2`, `val_size=0.1`, maximum citation context size 8, local-relevance neighbor sampling, and a 2-hop citation graph. The model used the `allenai/scibert_scivocab_uncased` tokenizer with text, metadata, and citation dimensions of 768, 256, and 256, respectively, a fusion dimension of 512, LoRA adaptation (`r=8`, `alpha=16`, dropout 0.1), two citation-transformer layers with four heads, latent-

graph reasoning enabled, and the backbone frozen up to layer 6. Training was performed in BF16 with batch size 64, one pretraining epoch, two fine-tuning epochs, learning rates of 2×10^{-5} and 1×10^{-5} for pretraining and fine-tuning, weight decay 0.01, maximum gradient norm 1.0, contrastive temperature 0.07, metadata negative weight 0.25, and pseudo-labeling configured with $\beta = 0.9$, warmup of 1 epoch, minimum 1 pseudo-label per class, EMA momentum 0.9, and distribution alignment enabled.

Metric	full	text_only	text_metadata	text_citation
Accuracy	0.612 (0.008)	0.548 (0.011)	0.571 (0.009)	0.594 (0.010)
Balanced accuracy	0.341 (0.013)	0.272 (0.015)	0.298 (0.012)	0.319 (0.014)
Micro-F1	0.612 (0.008)	0.548 (0.011)	0.571 (0.009)	0.594 (0.010)
Macro-F1	0.378 (0.012)	0.296 (0.014)	0.327 (0.011)	0.354 (0.013)
Weighted-F1	0.583 (0.009)	0.519 (0.012)	0.544 (0.010)	0.566 (0.011)
Macro precision	0.402 (0.014)	0.318 (0.016)	0.349 (0.013)	0.378 (0.014)
Macro recall	0.367 (0.011)	0.288 (0.013)	0.316 (0.011)	0.343 (0.012)
Matthews CC	0.581 (0.010)	0.512 (0.013)	0.539 (0.011)	0.563 (0.012)
Cohen's κ	0.578 (0.010)	0.508 (0.013)	0.535 (0.011)	0.560 (0.012)
Macro OvR AUROC	0.812 (0.006)	0.764 (0.009)	0.781 (0.007)	0.799 (0.007)
Log-loss	1.842 (0.031)	2.118 (0.042)	2.014 (0.037)	1.913 (0.034)

Metric	full	text	text+meta	text+cit
Time / epoch (min)	~18	~11	~13	~16
Peak VRAM (GB)	~22	~15	~17	~20

Table 3: OpenAlex-AI preliminary results, including classification performance and efficiency across the evaluated model variants.

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